

CHEATSHEET · QUICK REFERENCE

Lab 26 — Backup Strategy: Full, Synthetic, Incremental, Differential, Snapshot

Build full/incremental/differential/synthetic backups with tar and restic, compare media and GFS rotation, and validate every backup with a test restore.

Backup methods

```
tar -czf /srv/backup/full-$(date +%F).tgz -C /srv data
Full backup

tar --listed-incremental=/srv/backup/snap.snar -czf inc.tgz -C /srv data
Incremental (since last)

tar --listed-incremental=/srv/backup/snap-full.snar -czf diff.tgz -C /srv data
Differential (since last full)

restic -r /srv/backup/restic backup /srv/data
Synthetic full (dedup snapshot)
```

Method trade-offs

Type	Size	Restore complexity
Full	Largest	1 file
Incremental	Smallest	Full + all increments
Differential	Grows daily	Full + latest diff
Synthetic full	Delta only	1 file
Snapshot	COW delta	Instant (same host)

Restore methods + validation

```
tar -xzf full.tgz -C /srv
Overwrite restore

tar -xzf full.tgz -C /srv/restore-test
Side-by-side restore

restic restore latest --target /srv/restore-alt
Alternate-location restore

restic -r /srv/backup/restic check
Media integrity check
```

GFS rotation + media

Tier / medium	Detail
Son	Daily incrementals (7 kept)
Father	Weekly fulls (4 kept)
Grandfather	Monthly fulls (12 kept)
Tape / Disk / Cloud	Cheap+durable / fast / off-site